

$$\min_{P,Z,Q,E} \sum_{s=1}^S \|Z_s\|_* + \beta \sum_{s=1}^S \|Q_s\|_F^2 + \gamma \sum_{s=1}^S \|E_s\|_{2,1} + \delta \sum_{s=1}^S \|X_s - XQ_s\|_F^2 + \kappa \sum_{s=1}^S \|P_s X_s - P_s X_s Z_s\|_F^2 + \lambda \sum_{s \neq t} \text{HSIC}(Z_s, Z_t)$$

$$\text{s.t. } X_s = X_s Z_s + E_s, P_s^T P_s = I$$

其中:

$X = [X_1, X_2, \dots, X_S] \in \mathbb{R}^{D \times N}$, 是多模态数据, D 为原始数据的特征数, N 为所有模态数据的样本总数 $N = \sum^S n_s$

$X_s \in \mathbb{R}^{D \times n_s}$, 是每个模态的数据

$P_s \in \mathbb{R}^{d \times D}$, d 为降维之后特征数

$Q_s \in \mathbb{R}^{N \times n_s}$, n_s 为单个模态数据的样本数

$Z_s \in \mathbb{R}^{n_s \times n_s}$

$E_s \in \mathbb{R}^{D \times n_s}$

引入辅助变量 J

$$\min_{P,Z,Q,E,J} \sum_{s=1}^S \|J_s\|_* + \beta \sum_{s=1}^S \|Q_s\|_F^2 + \gamma \sum_{s=1}^S \|E_s\|_{2,1}$$

$$+ \delta \sum_{s=1}^S \|X_s - XQ_s\|_F^2 + \kappa \sum_{s=1}^S \|P_s X_s - P_s X_s Z_s\|_F^2$$

$$+ \lambda \sum_{s \neq t} \text{HSIC}(Z_s, Z_t)$$

$$\text{s.t. } X_s = X_s Z_s + E_s, P_s^T P_s = I, Z = J$$

↓ 构建增广拉格朗日函数

$$\mathcal{L}(Z, P, Q, E, J, \Gamma, \Delta) \sum_{s=1}^S \|J_s\|_* + \beta \sum_{s=1}^S \|Q_s\|_F^2 + \gamma \sum_{s=1}^S \|E_s\|_{2,1}$$

$$+ \delta \sum_{s=1}^S \|X_s - XQ_s\|_F^2 + \kappa \sum_{s=1}^S \|P_s X_s - P_s X_s Z_s\|_F^2$$

$$+ \lambda \sum_{s \neq t} \text{HSIC}(Z_s, Z_t)$$

$$+ \frac{\mu}{2} \sum_{s=1}^S \left(\left\| X_s - X_s Z_s - E_s + \frac{\Gamma}{\mu} \right\|_F^2 + \left\| Z_s - J_s + \frac{\Delta}{\mu} \right\|_F^2 \right)$$

一次固定一个模式

$$\mathcal{L}(Z, P, Q, E, J, \Gamma, \Delta) \|J_s\|_* + \beta \|Q_s\|_F^2 + \gamma \|E_s\|_{2,1}$$

$$+ \delta \|X_s - XQ_s\|_F^2 + \kappa \|P_s X_s - P_s X_s Z_s\|_F^2$$

$$+ \lambda \sum_{t=1, t \neq s}^S \text{HSIC}(Z_s, Z_t)$$

$$+ \frac{\mu}{2} \left(\left\| X_s - X_s Z_s - E_s + \frac{\Gamma}{\mu} \right\|_F^2 + \left\| Z_s - J_s + \frac{\Delta}{\mu} \right\|_F^2 \right)$$

HSIC约束选择内积核函数 $K_s = Z_s^T Z_s$

$$\sum_{t=1, t \neq s}^S \text{HSIC}(Z_s, Z_t) = \sum_{t=1, t \neq s}^S \text{Tr}(H K_s H K_t) = \sum_{t=1, t \neq s}^S \text{Tr}(Z_s H K_t H Z_s^T) = \text{Tr}(Z_s K Z_s^T)$$

其中:

$$K=\sum_{t=1,t\neq s}^SHK_tH$$

$$\begin{aligned} &\mathcal{L}(Z,P,Q,E,J,\Gamma,\Delta)\|J_s\|_*+\beta\|Q_s\|_F^2+\gamma\|E_s\|_{2,1}\\ &+\delta\|X_s-XQ_s\|_F^2+\kappa\|P_sX_s-P_sX_sZ_s\|_F^2\\ &+\lambda\text{Tr}(Z_sKZ_s^T)\\ &+\frac{\mu}{2}\left(\left\|X_s-X_sZ_s-E_s+\frac{\Gamma}{\mu}\right\|_F^2+\left\|Z_s-J_s+\frac{\Delta}{\mu}\right\|_F^2\right) \end{aligned}$$

其中：

$$K=\sum_{t=1,t\neq s}^SHK_tH$$

$$H=I-\frac{1}{n}11^T \text{是中心化矩阵,}$$

↴ 更新

step 1 J

$$J_{k+1}=\Theta(\frac{1}{\mu})\left(Z+\frac{\Gamma}{\mu}\right)$$

step 2 Z

$$\min_{Z_s}\kappa\|P_sX_s-P_sX_sZ_s\|_F^2+\lambda\text{Tr}(Z_sKZ_s^T)+\frac{\mu}{2}\left(\left\|X_s-X_sZ_s-E_s+\frac{\Gamma}{\mu}\right\|_F^2+\left\|Z_s-J_s+\frac{\Delta}{\mu}\right\|_F^2\right)$$

范数转化成tr的形式

$$\begin{aligned} &\min_{Z_s}\kappa\text{Tr}((P_sX_s-P_sX_sZ_s)(P_sX_s-P_sX_sZ_s)^T)+\lambda\text{Tr}(Z_sKZ_s^T)\\ &+\frac{\mu}{2}\left(\text{Tr}\left(\left(X_s-X_sZ_s-E_s+\frac{\Gamma}{\mu}\right)\left(X_s-X_sZ_s-E_s+\frac{\Gamma}{\mu}\right)^T\right)+\text{Tr}\left(\left(Z_s-J_s+\frac{\Delta}{\mu}\right)\left(Z_s-J_s+\frac{\Delta}{\mu}\right)^T\right)\right) \end{aligned}$$

求导

function:

$$\begin{aligned} f&=kappa\cdot\text{tr}((P\cdot X-P\cdot X\cdot Z)\cdot(P\cdot X-P\cdot X\cdot Z)^\top)+lambda\cdot\text{tr}(Z\cdot K\cdot Z^\top)\\ &+mu/2\cdot(\text{tr}((X-X\cdot Z-E+Lambda/2)\cdot(X-X\cdot Z-E+Lambda/2)^\top)+\text{tr}((Z-J+Delta/2)\cdot(Z-J+Delta/2)^\top)) \end{aligned}$$

gradient:

$$\begin{aligned} \frac{\partial f}{\partial Z}&=lambda\cdot Z\cdot K^\top-2\cdot kappa\cdot(P\cdot X)^\top\cdot(P\cdot X-P\cdot X\cdot Z)\\ &+lambda\cdot Z\cdot K-mu\cdot X^\top\cdot(X-X\cdot Z-E+1/2\cdot Lambda)+mu\cdot(Z-J+1/2\cdot Delta)\\ &=0 \end{aligned}$$

→

$$(2\kappa A^TA+\mu X^TX+\mu I)Z+Z\lambda(K^T+K)=2\kappa A^TA+\mu X^TX-\mu X^T(E-\Lambda/2)+\mu J-\mu/2\Delta$$

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z = sylvester(A, B, C);
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step 3 P

$$\begin{aligned} & \min_{P_s} \kappa \|P_s X_s - P_s X_s Z_s\|_F^2 \\ &= \min_{P_s} \kappa \text{Tr}(PX(Z-I)(Z-I)^T X^T P^T) \end{aligned}$$

转化为求特征值

$$X(\kappa(Z-I)(Z-I))X^T p = \xi p$$

选前lower_dim个向量拼接成 P_s^T

step 4 Q

$$\min_{Q_s} \beta \|Q_s\|_F^2 + \delta \|X_s - XQ_s\|_F^2$$

求导=0

$$Q_s = (\beta I + \delta X^T X)^{-1} \delta X^T X_s$$

step E

$$\min_{E_s} \gamma \|E_s\|_{2,1} + \frac{\mu}{2} \left\| X_s - X_s Z_s - E_s + \frac{\Gamma}{\mu} \right\|_F^2$$

$$V = X_s - X_s Z_s + \frac{\Gamma}{\mu}$$

$$[E^*]_{:,i} = \begin{cases} [V_{:,i}]_2 - \alpha & \text{if } \|V_{:,i}\|_2 > \alpha \\ 0 & \text{otherwise} \end{cases} \tag{16}$$

where $[E^*]_{:,i}$ is the column of the optimal solution E^* .